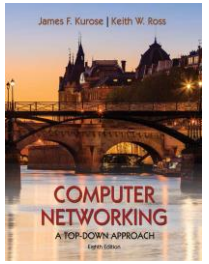
AinShams University
Faculty of Engineering

CSE 351: Computer networks
Section 4

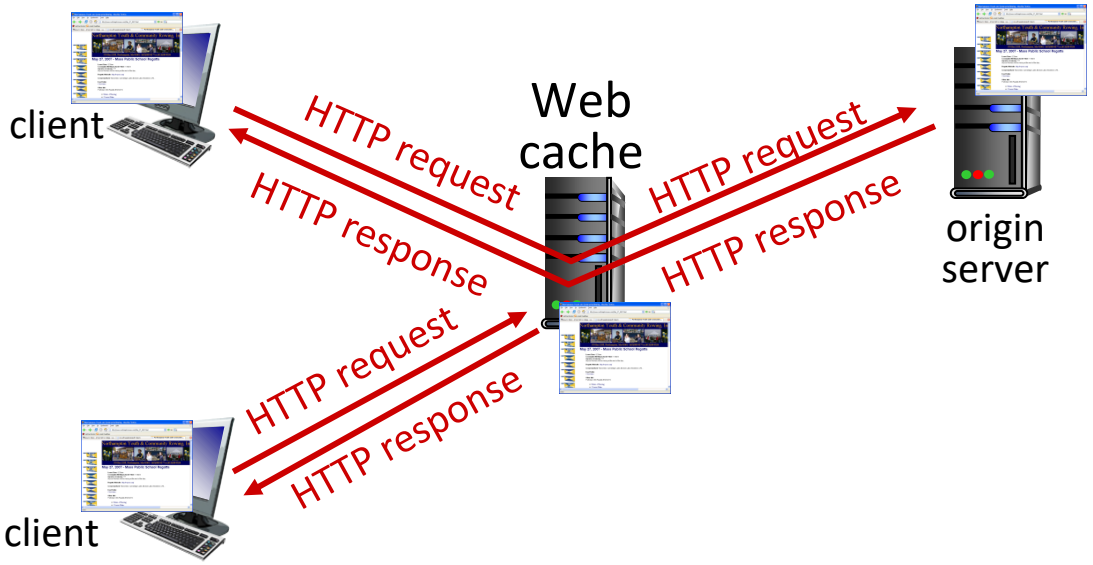
Eng. Noha Wahdan

Class textbook:
Computer Networking: A Top-Down Approach (8th ed.)
J.F. Kurose, K.W. Ross
Pearson, 2020
http://gaia.cs.umass.edu/kurose_ross



Web caches (aka proxy servers)

Goal: satisfy client requests without involving origin server



- Web cache acts as both client and server
  - **server** for original requesting client
  - **client** to origin server

Why Web caching?

- reduce response time for client request (cache is closer to client)
- reduce traffic on an institution's access link
- enables "poor" content providers to more effectively deliver content

- server tells cache about *which object to cache* (in response header):

```
Cache-Control: max-age=<seconds>
```

```
Cache-Control: no-cache
```

Recall : The Expected Value

$$\mathbf{E}[X] = \sum_{i=1}^{k} x_i\, p_i = x_1 p_1 + x_2 p_2 + \cdots + x_k p_k.$$

- As an example, suppose there is a 20% chance of 1 inch of rain, a 70% chance of 2 inches of rain, and a 10% chance of 3 inches of rain.

Amount of Rain	Probability
1 inch	0.2
2 inches	0.7
3 inches	0.1

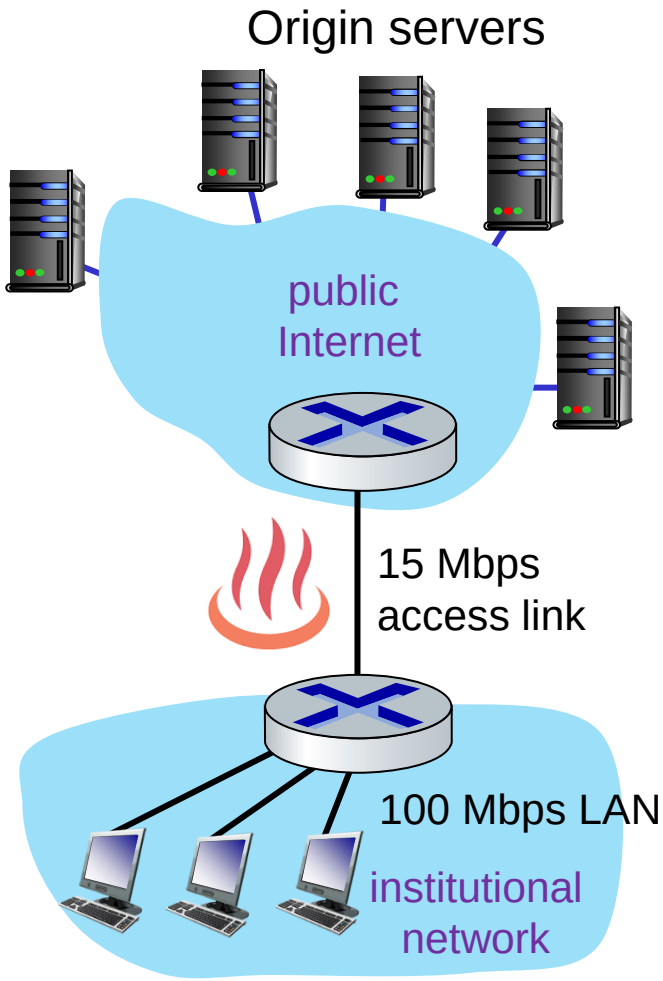
- We would calculate the expected value for the amount of rain to be:
- Expected value = 0.2*1 + 0.7*2 + 0.1*3 = 1.9 inches

Sheet 2: Question 14

In the shown figure, there is an institutional network connected to the Internet. Suppose that the average object size is 850,000 bits and that the average request rate from the institution's browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average. Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay. For the average access delay, use $\Delta/(1 - \Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the arrival rate of objects to the access link.

a. Find the total average response time.

b. Now suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.4. Find the total response time.



Sheet 2 Q14: (a) without a Web Cache

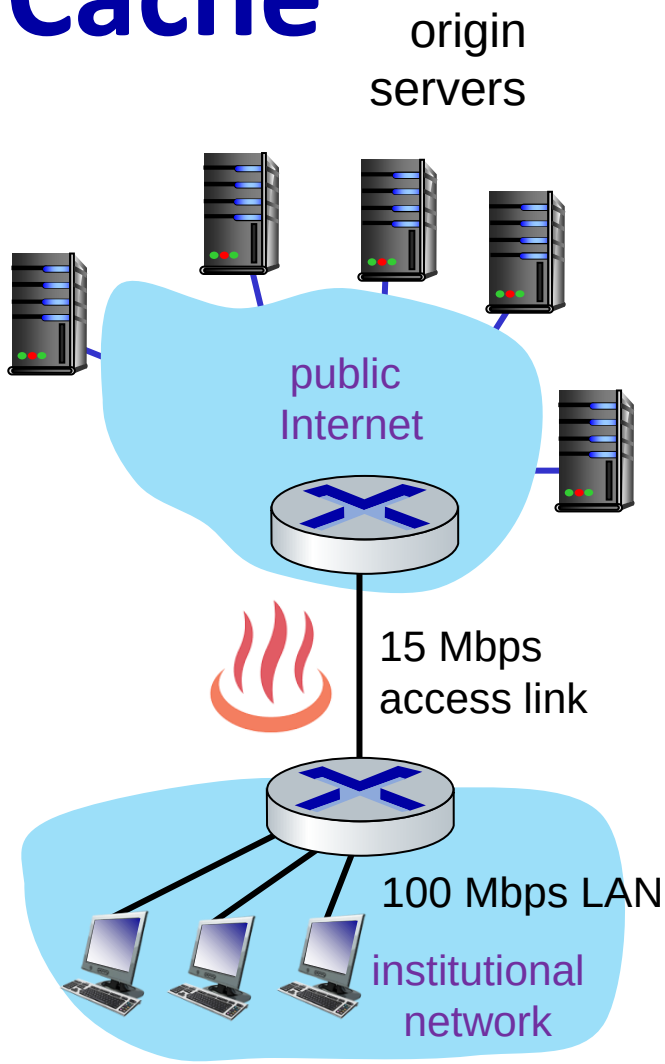
- Avg. amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response : 3 sec
- Avg. object size: 850,000 bits
- average request rate from browsers to origin servers: 16 req/sec

Access Link Performance:

- Avg. Transmission Time Δ = (850,000 bits)/(15 x 10^6 bits/sec) = 0.0567 sec
- Avg. arrival rate β = 16 req/sec
- Avg. delay = $\Delta/(1 - \Delta\beta)$ = (0.0567)/(1- 0.0567 x 16) ≈ 0.6 sec

Total avg. response time= 0.6 sec + 3 sec = 3.6 sec.

Note that delay on LAN is ignored because it is < 0.01 sec



Sheet 2 Q14 : (b) with a Web Cache

Solutions:

- Buy a faster access link →*Cost:* expensive!
- Install a web cache →*Cost:* cheap!

suppose a cache is installed and the miss rate is 0.4

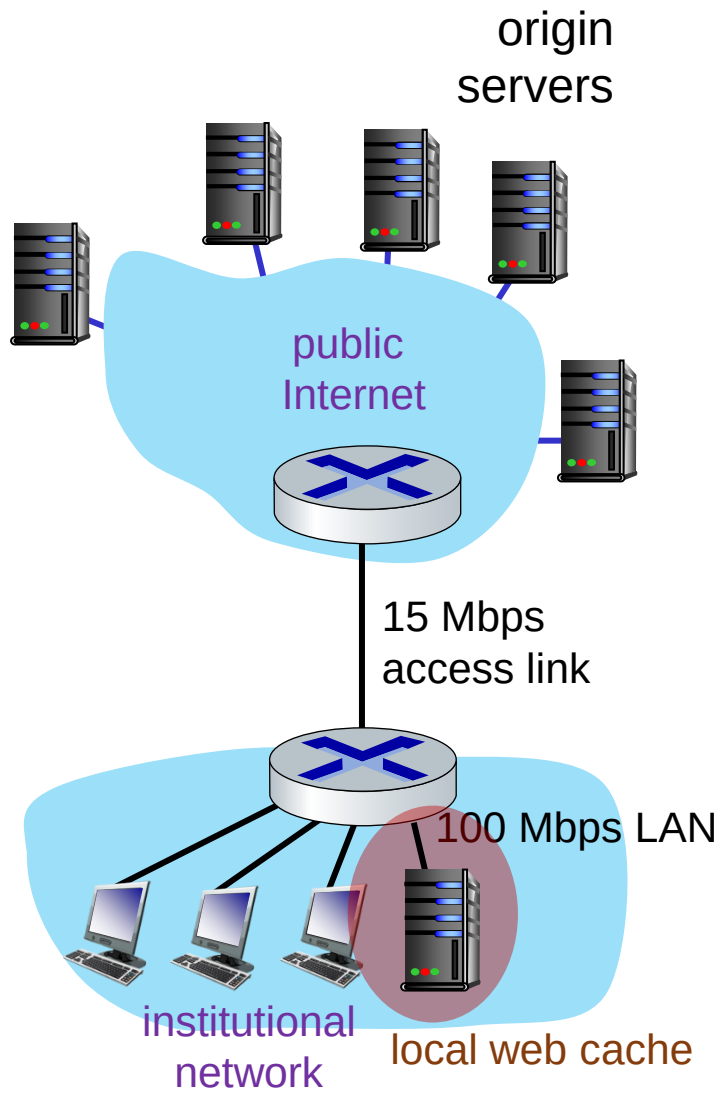
Access Link Performance:

- Avg. Transmission Time Δ = (850,000 bits)/(15 x 10^6 bits/sec) = 0.0567 sec
- Avg. arrival rate β = 0.4 x 16 = 6.4 req/sec
- Avg. delay = Δ/(1 – Δβ) = (0.0567)/(1- 0.0567 x 6.4) ≈ 0.089 sec

Reduced to ≈one third

Total avg. response time = 0.6x(0)+0.4x(0.089+3) = 1.24 sec.

Web Caching: reduces response time for client request and reduces traffic on the access link



Sheet 2: Question 5

Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

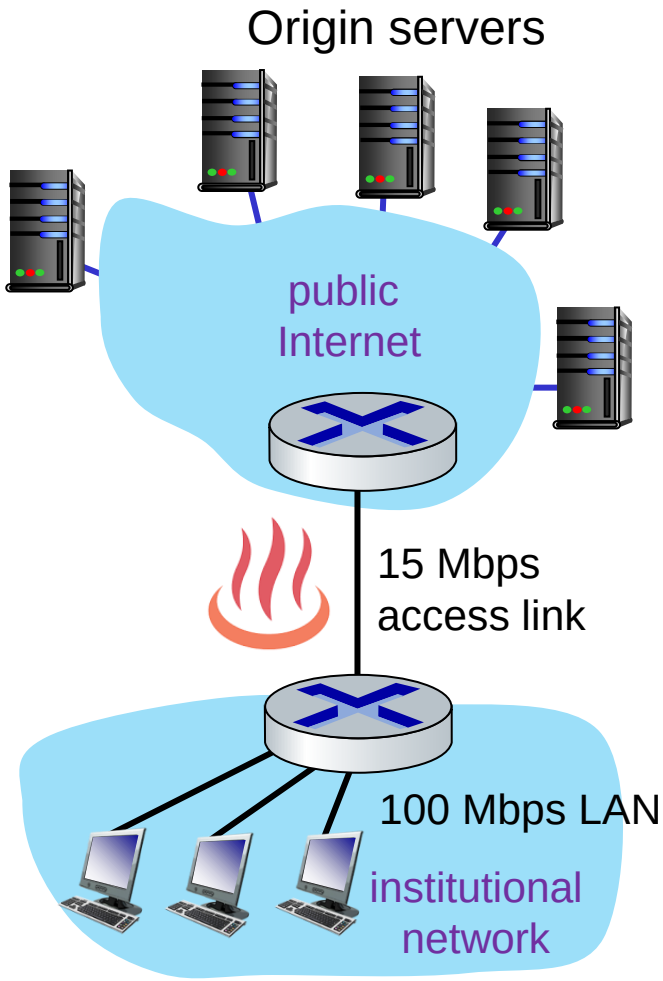
Web caching can bring the desired content “closer” to the user, perhaps to the same LAN to which the user host is connected.
Web caching can reduce the delay for all objects, even objects that are not cached, since caching reduces the traffic on links.

Sheet 3: Question 14

In the shown figure, there is an institutional network connected to the Internet. Suppose that the average object size is 1,000,000 bits and that the average request rate from the institution's browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average. Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay. For the average access delay, use $\Delta/(1 - \Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the arrival rate of objects to the access link.

a. Find the total average response time.

b. Now suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.4. Find the total response time.



Conditional GET

Goal: don't send object if cache has up-to-date cached version

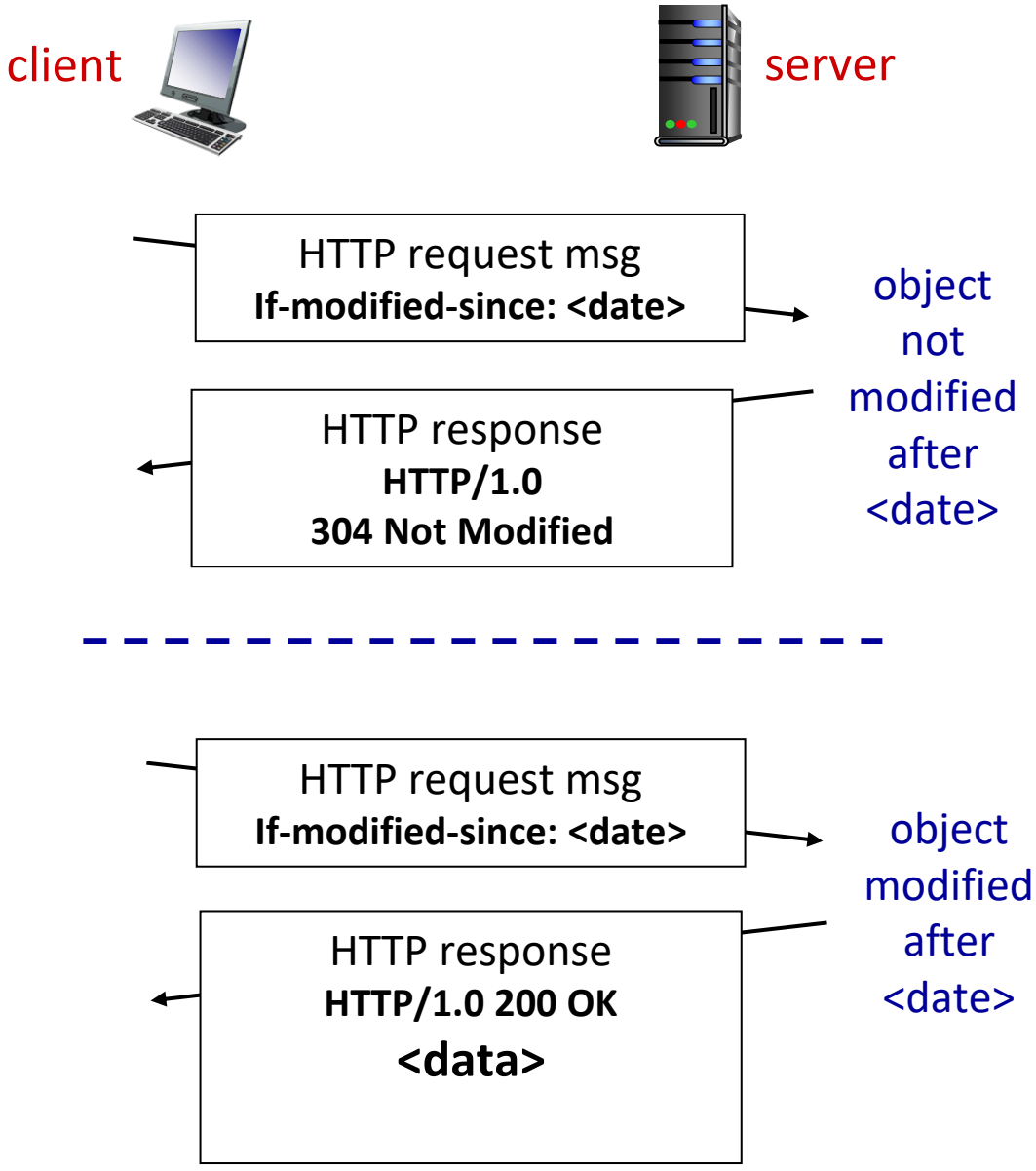
- no object transmission delay (or use of network resources)

- *client:* specify date of cached copy in HTTP request

  If-modified-since: <date>

- *server:* response contains no object if cached copy is up-to-date:

  HTTP/1.0 304 Not Modified



Domain Name System (DNS)

- An application-layer protocol translates web server's name to its corresponding IP address.
- A distributed database implemented in a hierarchy of DNS servers
- DNS protocol runs over UDP and uses port 53

DNS used by App layer protocols

What Happens when a browser requests www.someschool.edu/index.html.

1. The same user machine runs the client side of the DNS application.

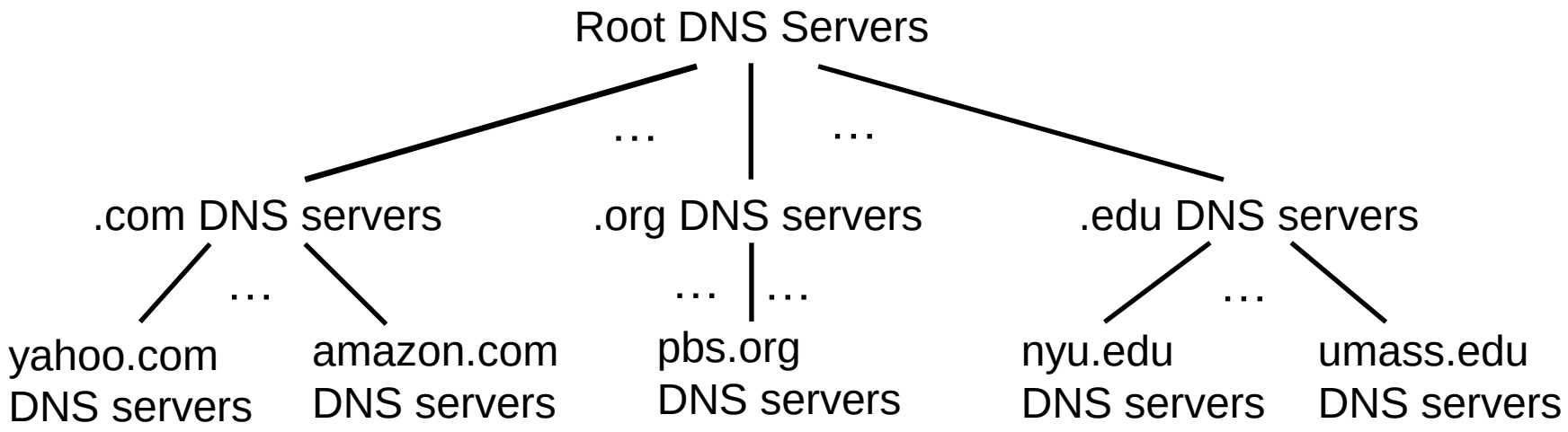
2. The browser extracts the hostname, www.someschool.edu, from the URL and passes the hostname to the client side of the DNS application.

3. The DNS client sends a query containing the hostname to a DNS server.

4. The DNS client eventually receives a reply, which includes the IP address for the hostname.

5. Once the browser receives the IP address from DNS, it can initiate a TCP connection to the HTTP server process located at port 80 at that IP address.

DNS: a distributed, hierarchical database



Root

Top Level Domain

Authoritative

Client wants IP address for www.amazon.com; 1st approximation:

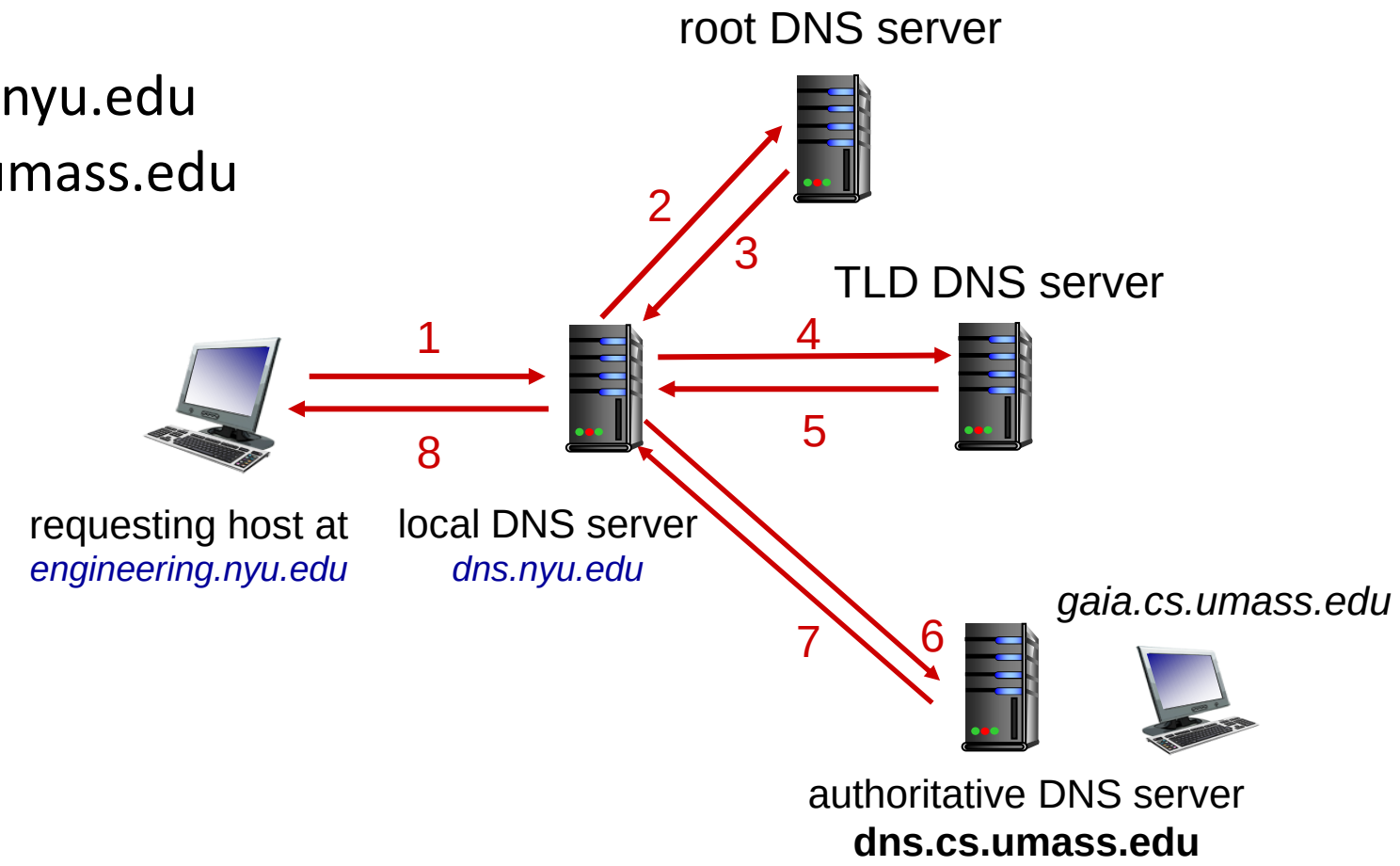
- client queries root server to find .com DNS server
- client queries .com DNS server to get amazon.com DNS server
- client queries amazon.com DNS server to get IP address for www.amazon.com

DNS name resolution: iterated query

Example: host at engineering.nyu.edu wants IP address for gaia.cs.umass.edu

Iterated query:

- contacted server replies with name of server to contact
- "I don't know this name, but ask this server"



Caching DNS Information

- Once (any) name server learns mapping, it *caches* mapping, and *immediately* returns a cached mapping in response to a query
- Advantages?
- Disadvantages?

DNS records

DNS: distributed database storing resource records (RR)

RR format: (`name, value, type, ttl`)

type=A

- `name` is hostname
- `value` is IP address

type=NS

- `name` is domain (e.g., foo.com)
- `value` is hostname of authoritative name server for this domain

type=CNAME

- `name` is alias name for some "canonical" (the real) name
- www.ibm.com is really servereast.backup2.ibm.com
- `value` is canonical name

type=MX

- `value` is name of SMTP mail server associated with `name`

Sheet 2: Question 7

Is it possible for an organization's Web server and mail server to have the same alias for a hostname (for example, foo.com)? What would be the type for the RR that contains the hostname of the mail server?

Yes, an organization's mail server and Web server can have the same alias for a hostname.
The MX record is used to map the mail server's hostname to its IP address.

Sheet 2: Question 18

Suppose you can access the caches in the local DNS servers of your department. Can you propose a way to roughly determine the Web servers (outside your department) that are most popular among the users in your department? Explain.

We can periodically take a snapshot of the DNS caches in those local DNS servers. The Web server that appears most frequently in the DNS caches is the most popular server.
This is because if more users are interested in a Web server, then DNS requests for that server are more frequently sent by users. Thus, that Web server will appear in the DNS caches more frequently.

Sheet 2: Question 19

Suppose that your department has a local DNS server for all computers in the department. You are an ordinary user (i.e., not a network/system administrator). Can you determine if an external Web site was likely accessed from a computer in your department a couple of seconds ago? Explain.

Yes, we can use dig to query that Web site in the local DNS server.
For example, “dig cnn.com” will return the query time for finding cnn.com. If cnn.com was just accessed a couple of seconds ago, an entry for cnn.com is cached in the local DNS cache, so the query time is very small (~0msec). Otherwise, the query time is large.
Hint: On Windows, you can use the command:
powershell "Measure-Command { nslookup www.IndiaMART.com}"

Sheet 2: Question 10

Consider an HTTP client that wants to retrieve a Web document at a given URL. The IP address of the HTTP server is initially unknown. What transport and application-layer protocols besides HTTP are needed in this scenario?

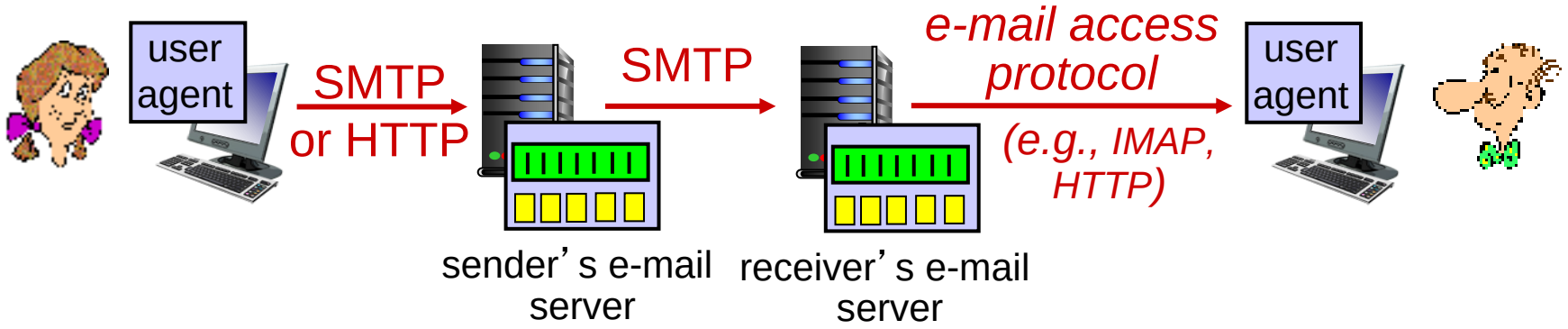
- Application layer protocols: DNS and HTTP.
- Transport layer protocols: UDP for DNS; TCP for HTTP.

Sheet 2: Question 13

Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of RTT1, . . . , RTTn. Further suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let RTT0 denote the RTT between the local host and the server containing the object. Assuming zero transmission time of the object, how much time elapses from when the client clicks on the link until the client receives the object?

- The total amount of time to get the IP address is RTT1 + RTT2 + ….. + RTTn .
- Once the IP address is known, RTT0 elapses to set up the TCP connection and another RTT0 elapses to request and receive the small object.
- The total response time is:
- 2 RTT0 + RTT1 + RTT2 + ….. + RTTn .

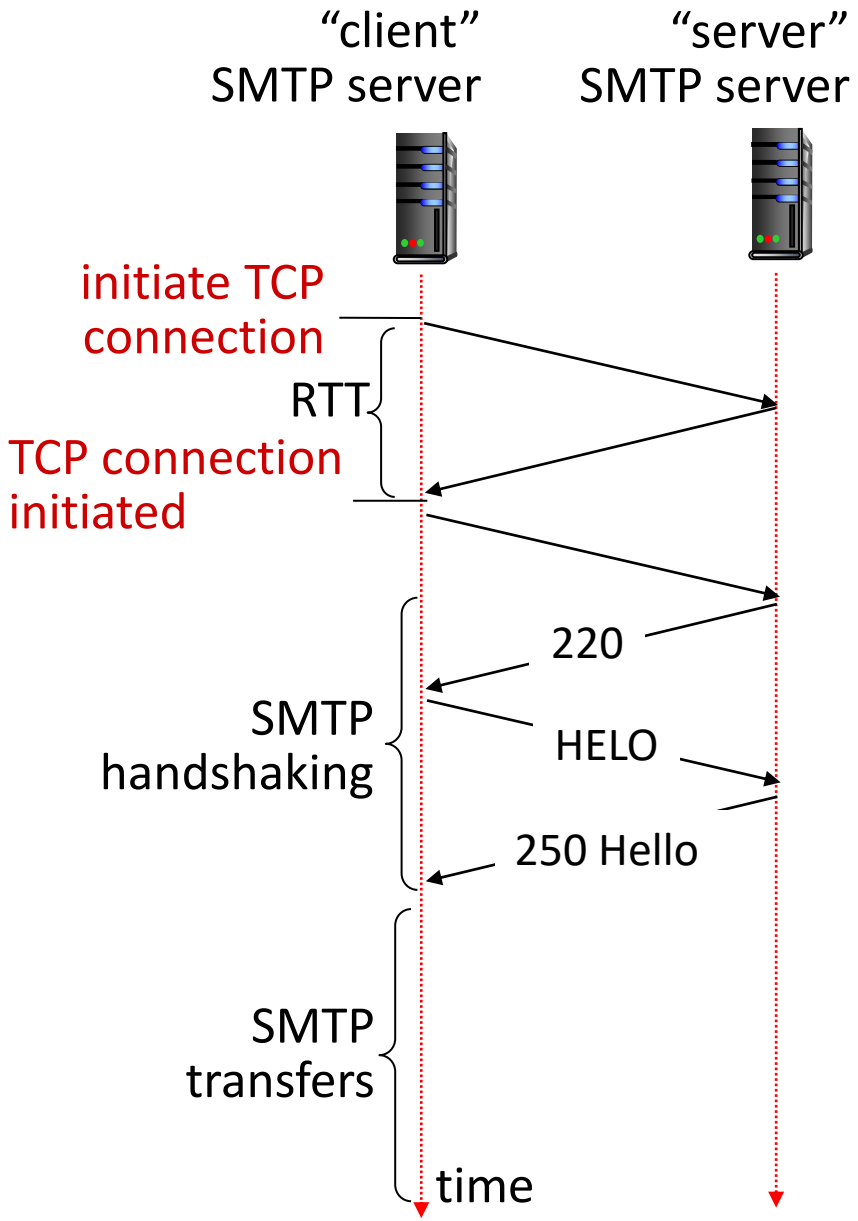
Mail access protocols



- SMTP: delivery/storage of e-mail messages to receiver's server
- mail access protocol: retrieval from server
  - IMAP: Internet Mail Access Protocol [RFC 3501]: messages stored on server, IMAP provides retrieval, deletion, folders of stored messages on server
- HTTP: gmail, Hotmail, Yahoo! Mail, etc. provides web-based interface on top of STMP (to send), IMAP (or POP) to retrieve e-mail messages

SMTP RFC (5321)

- uses TCP to reliably transfer email message from client (mail server initiating connection) to server, port 25
  - direct transfer: sending server (acting like client) to receiving server
- three phases of transfer
  - SMTP handshaking (greeting)
  - SMTP transfer of messages
  - SMTP closure
- command/response interaction (like HTTP)
  - commands: ASCII text
  - response: status code and phrase



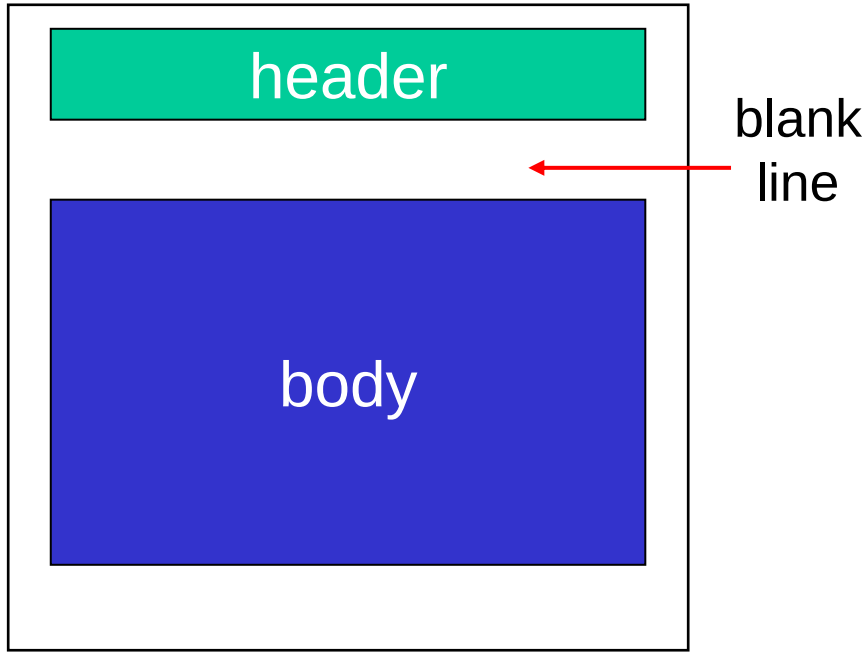
Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

Mail message format

- header lines, e.g.,
  - To:
  - From:
  - Subject:

  these lines, within the body of the email message area different from SMTP MAIL FROM:, RCPT TO: commands!
- Body: the “message” , ASCII characters only



Sheet 2: Question 16

What is the difference between MAIL FROM: in SMTP and From: in the mail message itself?

The MAIL FROM: in SMTP is a command from the SMTP client that identifies the sender of the mail message to the SMTP server.
The From: on the mail message itself is NOT an SMTP command, but rather is just a line in the body of the mail message

Sheet 2: Question 17

How does SMTP mark the end of a message body?
How about HTTP?
Can HTTP use the same method as SMTP to mark the end of a message body?
Explain.

- SMTP uses a line containing only a period to mark the end of a message body.
- HTTP uses “Content-Length header field” to indicate the length of a message body.
- No, HTTP cannot use the method used by SMTP, because HTTP message could be binary data, whereas in SMTP, the message body must be in 7-bit ASCII format.

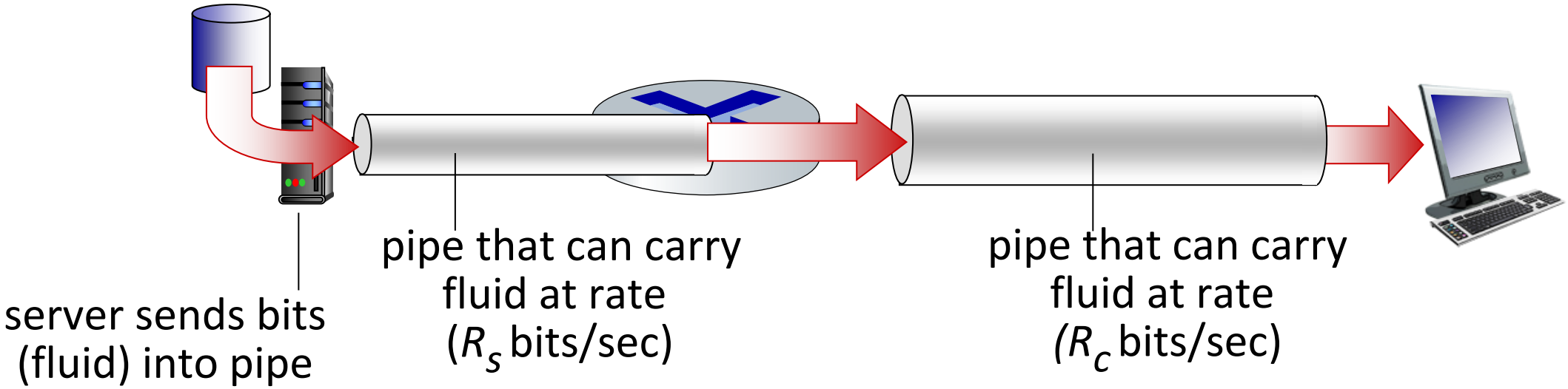
Sheet 2: Question 6

Suppose **Alice, with a Web-based e-mail account** (such as Hotmail or Gmail), sends a message to **Bob, who accesses his mail from his mail server using IMAP**. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.

The message is first sent from Alice's host to her mail server over HTTP. Alice's mail server then sends the message to Bob's mail server over SMTP. Bob then transfers the message from his mail server to his host over IMAP.

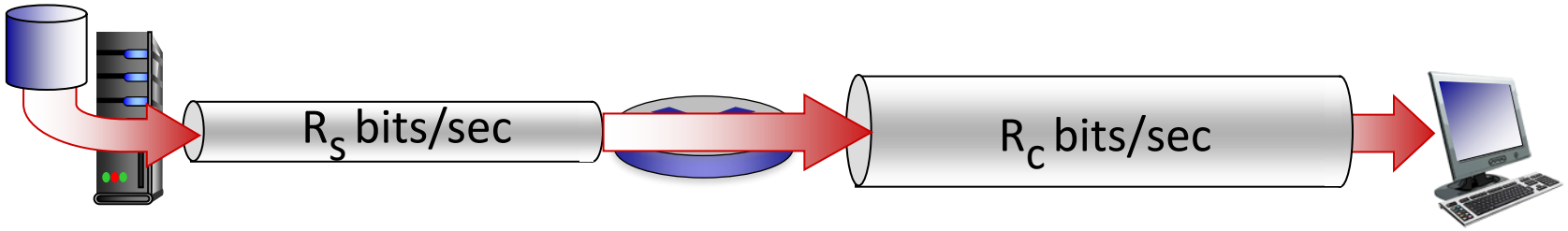
Throughput

- *End-to-end throughput:* rate (bits/time unit) at which bits are being delivered from sender to receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time

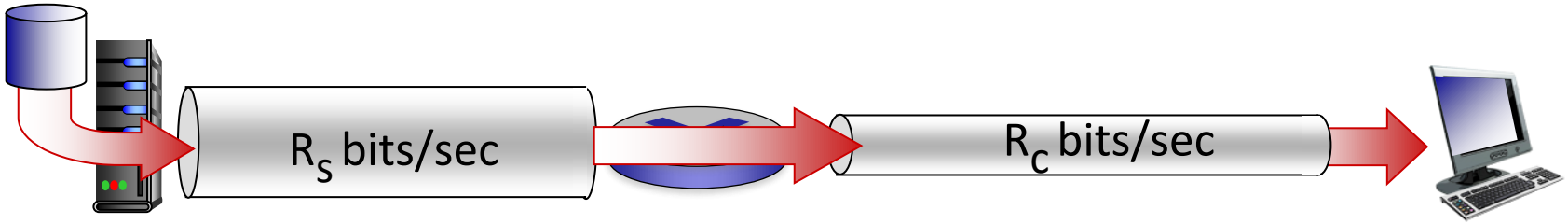


Throughput

$R_s < R_c$ What is average end-end throughput? R_s



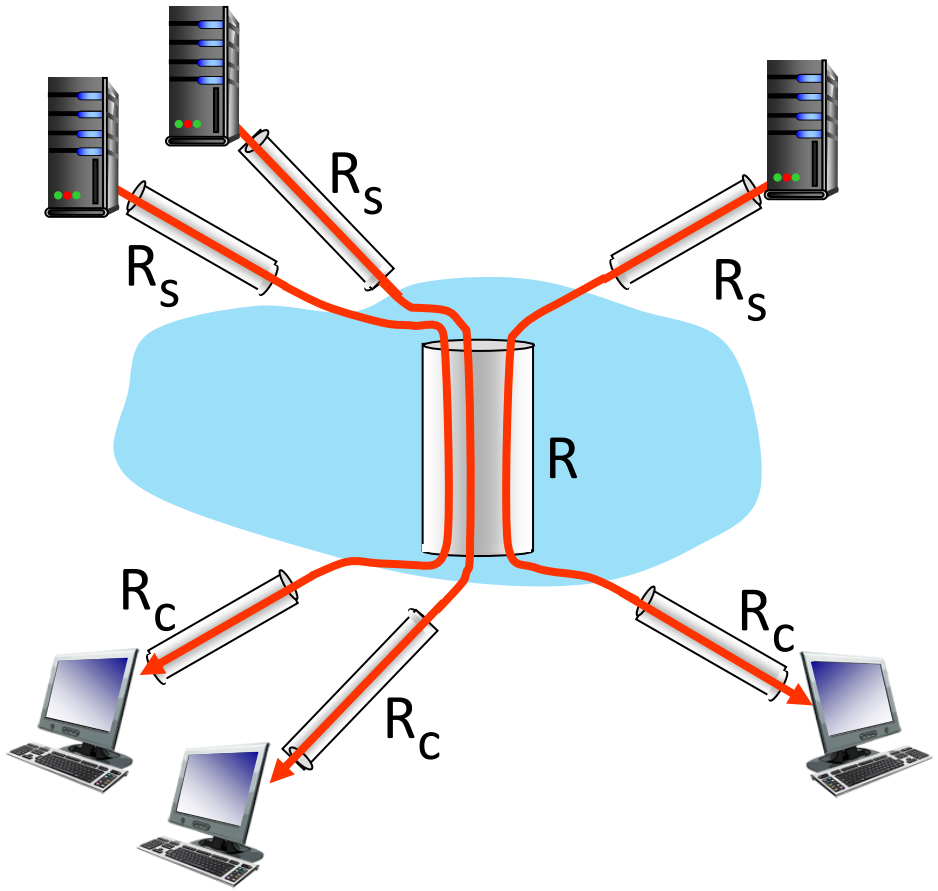
$R_s > R_c$ What is average end-end throughput? R_c



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: network scenario



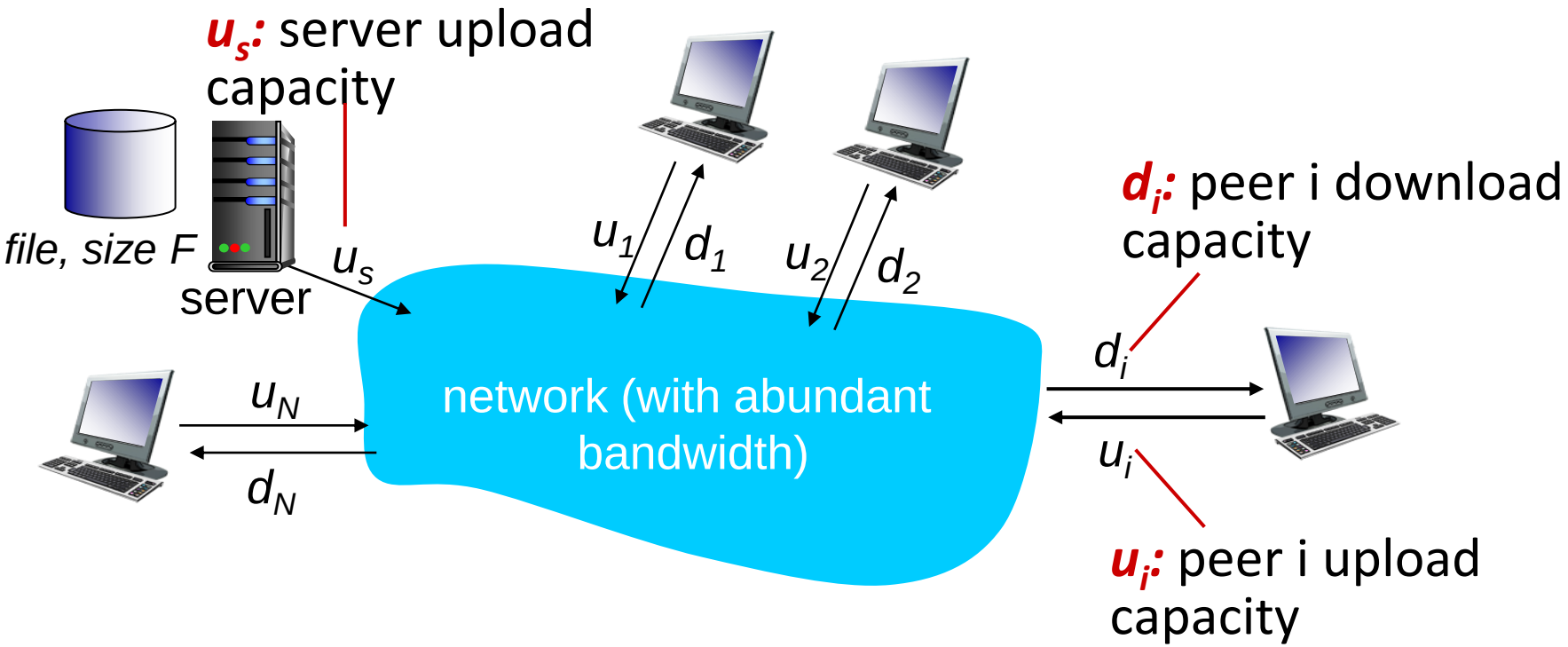
10 connections (fairly) share backbone bottleneck link *R* bits/sec

- per-connection end-end throughput: *min(R_c,R_s,R/10)*
- in practice: R_c or R_s is often bottleneck

File distribution: client-server vs P2P

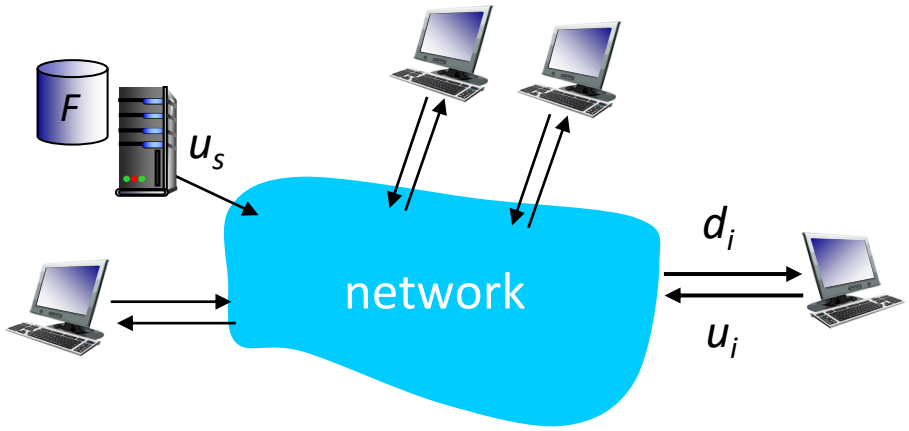
Q: how much time to distribute file (size *F*) from one server to *N peers*?

- peer upload/download capacity is limited resource



File distribution time: client-server

- *server transmission:* must sequentially send (upload) *N* file copies:
  - time to send one copy: F/u_s
  - time to send *N* copies: NF/u_s
- *client:* each client must download file copy
  - d_{min} = min client download rate
  - min client download time: F/d_{min}



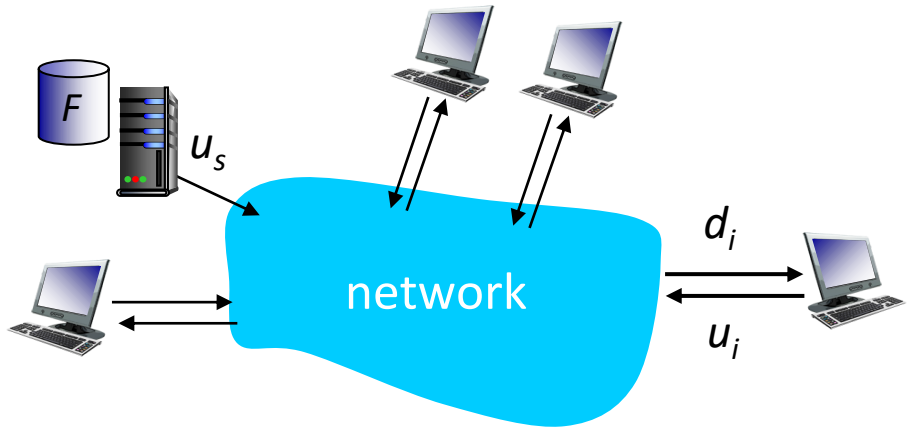
time to distribute F to N clients using client-server approach

$$D_{c\text{-}s} = max\{NF/u_s, F/d_{min}\}$$

increases linearly in N

File distribution time: P2P

- *server transmission:* must upload at least one copy:
  - time to send one copy: F/u_s
- *client:* each client must download file copy
  - min client download time: F/d_{min}
- *clients:* as aggregate must download NF bits
  - max upload rate (limiting max download rate) is $u_s + \Sigma u_i$



time to distribute F to N clients using P2P approach

$$D_{P2P} = max\{F/u_s, F/d_{min}, NF/(u_s + \Sigma u_i)\}$$

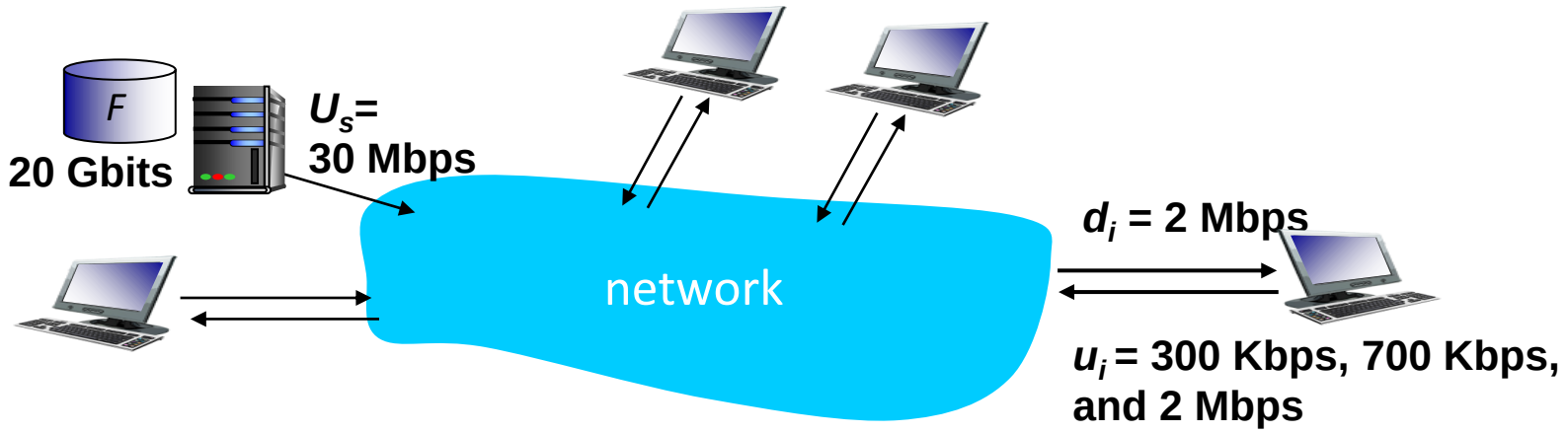
increases linearly in N …
… but so does this, as each peer brings service capacity

Sheet 2: Question 20

Consider distributing a file of *F* = 20 Gbits to *N* peers. The server has an upload rate of u_s = 30 Mbps, and each peer has a download rate of *di* = 2 Mbps and an upload rate of *u*. For *N* = 10, 100, and 1000 and *u* = 300 Kbps, 700 Kbps, and 2 Mbps, prepare a chart giving the minimum distribution time for each of the combinations of *N* and *u* for both client-server distribution and P2P distribution.

$$D_{cs} = max\ \{NF/u_s,\ F/d_{min}\}$$

$$D_{P2P} = max\{F/u_s,\ F/d_{min},\ NF/(u_s + \sum_{i=1}^{N} u_i)\}$$

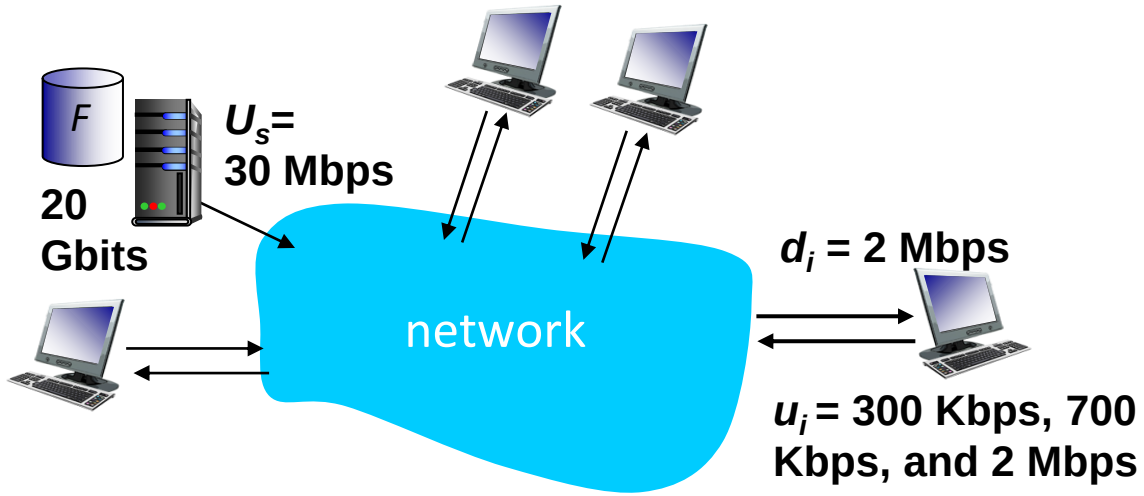


Sheet 2: Question 20 - client-server

u	N					
	10		100		1000	
	NF/u_s	F/d_{min}	NF/u_s	F/d_{min}	NF/u_s	F/d_{min}
300 kbps	6,666.67	10,000	66,666.67	10,000	666,666.67	10,000
700 kbps	6,666.67	10,000	66,666.67	10,000	666,666.67	10,000
2 Mbps	6,666.67	10,000	66,666.67	10,000	666,666.67	10,000

time to distribute F to N clients using client-server approach

$$D_{c\text{-}s} = max\{NF/u_s, F/d_{min}\}$$

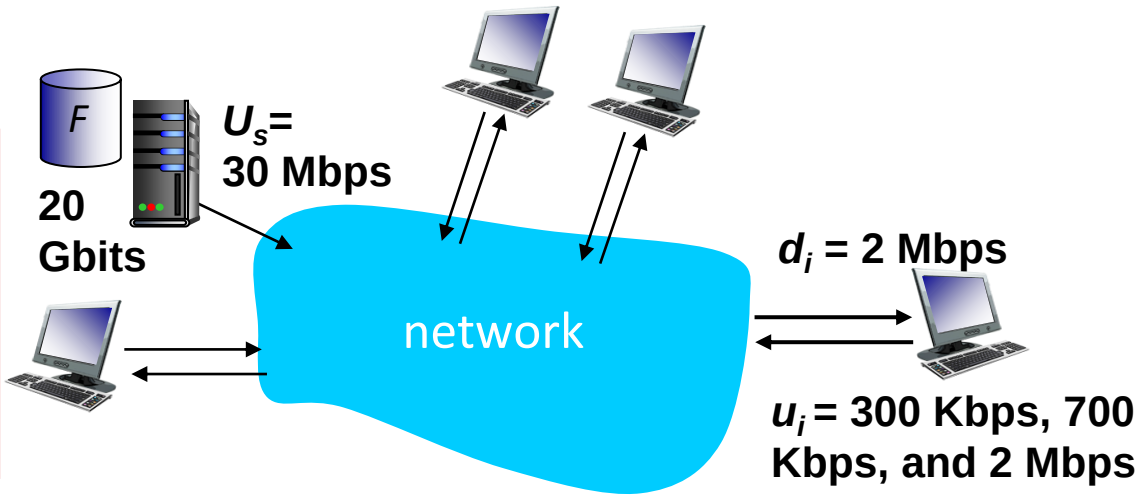


Sheet 2: Question 20 – P2P

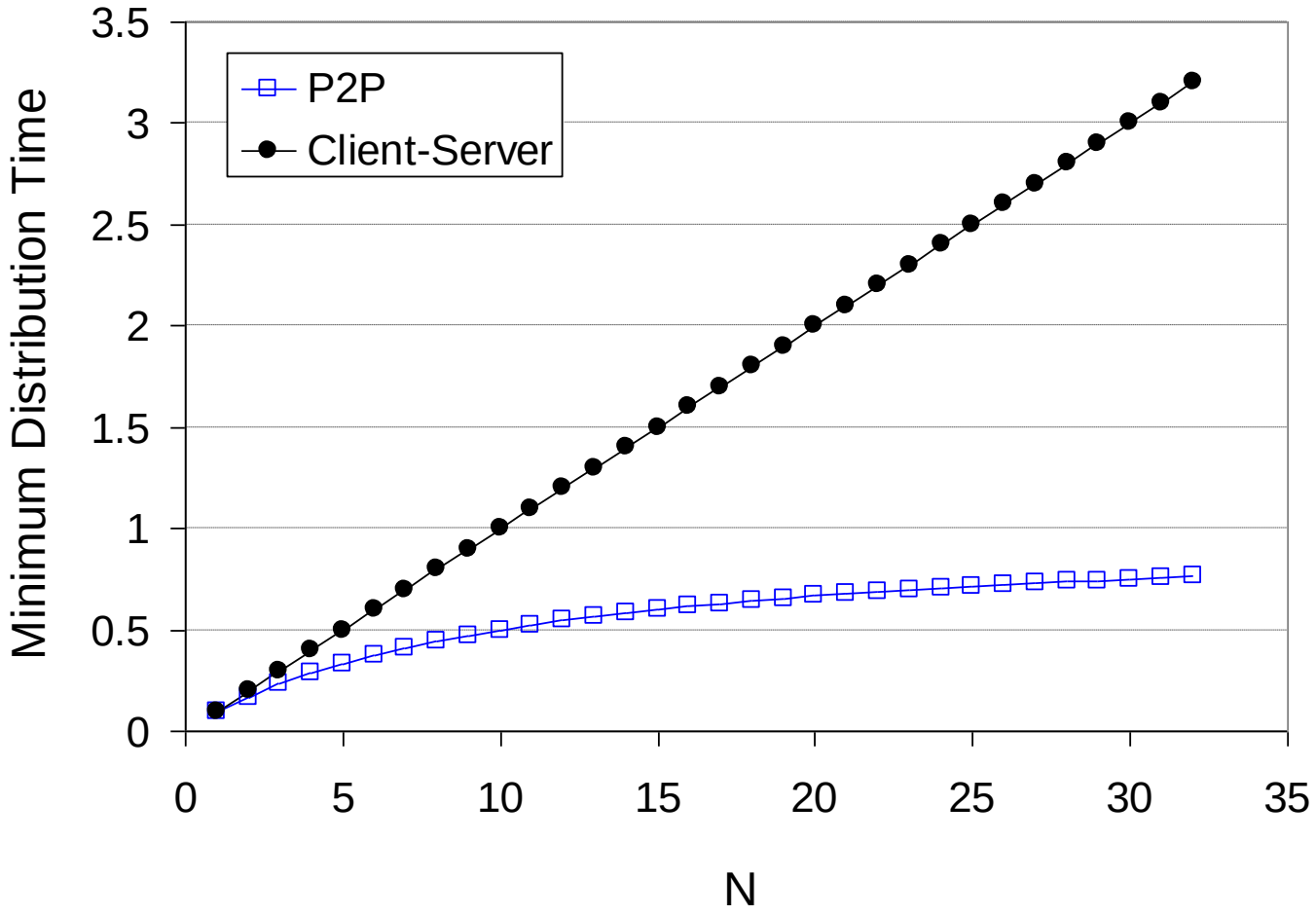
u	N								
	10			100			1000		
	F/u_s	F/d_{min}	$NF/(u_s + \Sigma u_i)$	F/u_s	F/d_{min}	$NF/(u_s + \Sigma u_i)$	F/u_s	F/d_{min}	$NF/(u_s + \Sigma u_i)$
300 kbps	666.67	10,000	6,060.60	666.67	10,000	33,333.33	666.67	10,000	60,606.06
700 kbps	666.67	10,000	5,405.41	666.67	10,000	20,000	666.67	10,000	27,397.26
2 Mbps	666.67	10,000	4,000	666.67	10,000	8,695.65	666.67	10,000	9,852.22

time to distribute F to N clients using P2P approach

$$D_{P2P} = max\{F/u_s, F/d_{min}, NF/(u_s + \Sigma u_i)\}$$



Client-server vs. P2P: example

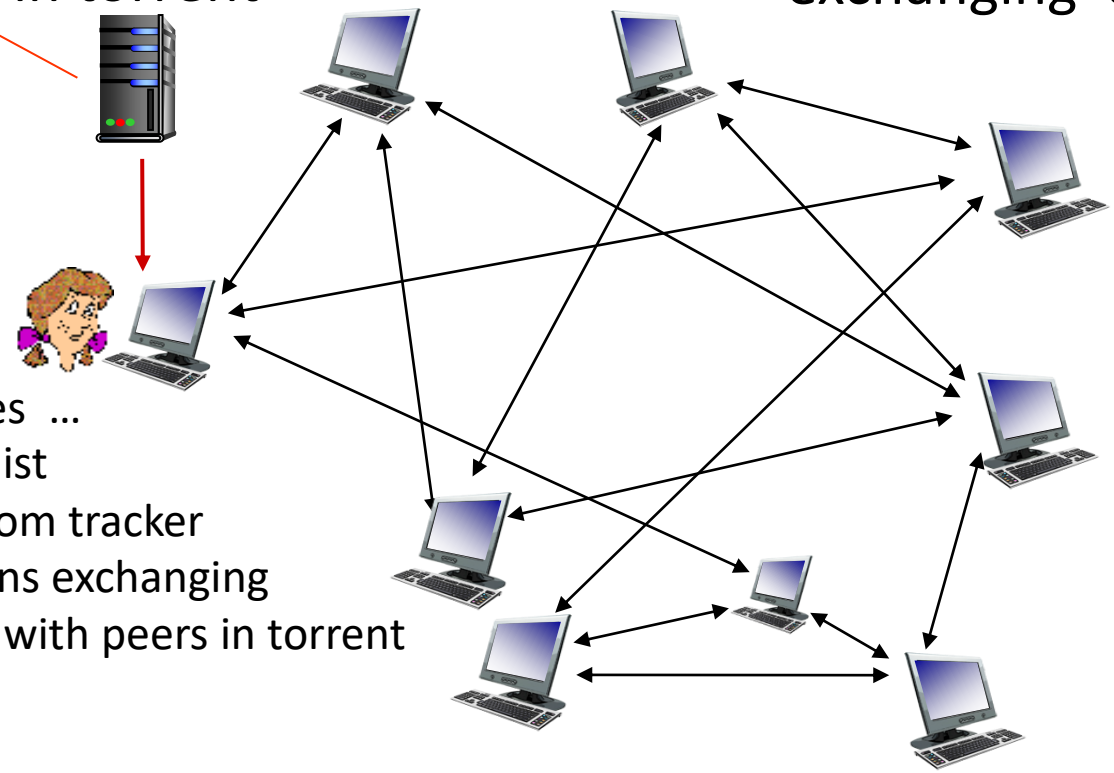


P2P file distribution: BitTorrent

- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



Alice arrives …
… obtains list
of peers from tracker
… and begins exchanging
file chunks with peers in torrent

Sheet 2

At any given instant of time, a peer in a torrent has a subset of chunks and will know which chunks its neighbors have.

- With this information, how does it decide which chunks it should request first from its neighbors?
- Rarest first

- To which of its neighbors should it send requested chunks?
- 4 top priority peers (Calculated every 10 seconds) + 1 random peer optimistically unchoked (every 30 seconds).

Sheet 2: Question 8

Consider a new peer Alice that joins BitTorrent without possessing any chunks. Without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk?

She will get her first chunk as a result of her being selected by one of her neighbors as a result of an "optimistic un-choke" for sending out chunks to her. According to BitTorrent mechanism, Alice may obtain first chunk from other peers which randomly select top four up-loaders of the member's peer list every 30 seconds

Sheet 2: Question 22

Suppose Bob joins a BitTorrent torrent, but he does not want to upload any data to any other peers (so called free-riding).

a. Bob claims that he can receive a complete copy of the file that is shared by the swarm. Is Bob's claim possible? Why or why not?

Yes. His first claim is possible, as long as there are enough peers staying in the swarm for a long enough time. Bob can always receive data through optimistic unchoking by other peers.

Sheet 2: Question 22 (Continued)

Suppose Bob joins a BitTorrent torrent, but he does not want to upload any data to any other peers (so called free-riding).

b. Bob further claims that he can further make his “free-riding” more efficient by using a collection of multiple computers (with distinct IP addresses) in the computer lab in his department. How can he do that?

His second claim is also true. He can run a client on each host, let each client “free-ride,” and combine the collected chunks from the different hosts into a single file. He can even write a small scheduling program to make the different hosts ask for different chunks of the file.

Thanks